

Lesson 3

Equipment: Raspberry Pi Python code editor, provided sheets, pens/pencils, post-it notes (or similar such as scrap paper etc.)

Summary

- Lesson 3 will involve pupils being introduced to variables. Pupils will first explore the motivation for variables through an unplugged activity before going on to see examples of variables being used in short programs.

Lesson Aims

- I can consider the meaning of the word “variable” and how it applies to our own use of language and sentence structure
- I can understand that variables are used to store information in a program
- I can recognise examples of variables potentially being used in modern applications, in particular in applications that use a “Hello {user}” welcome message

Curriculum Integration

- TCH 2-13a: pupils should understand that they have constructed a model of a physical variable through the process of exchanging values with another team.
- TCH 2-14a: pupils should be able to give a description of what a variable is and why we use variables in programming.
- TCH 2-15a: pupils should be able to implement a variable in their own short programs.

Plan

Starter (10 minutes)

- Prompt for pupils: Think about what the word “variable” means to you. What sort of things are “variable”? What stays the same?

Unplugged activity – Sentence Building (20 minutes)

- Pupils should get into teams of at least 3*.
- Pupils are all given a post-it note with either X, Y or Z on it with each team having at least one of X, Y and Z.
- Pupils are given the following sentence: “X went to the shop and bought Y and Z” and should write a value on the opposite side of their post-it note that could be substituted for their letter.
- Once all pupils have written a value, in their teams they should substitute all X, Y and Z values into the given sentence and note what sentence has been created
- After this, teams will swap with their set of post-it notes with another team. They should note the sentence that can be created with their new set of post-it notes.

- Pupils should reflect on whether the new sentence they have constructed is valid. They should be led to the understanding that X, Y and Z can be seen as variables.
- *if there are teams of more than 3 then pupils will have multiple X, Y or Z values. Pupils should try substituting in their different values for these. For the swapping exercise, teams should choose just 1 set of X, Y and Z values.

Taught Section (10 minutes)

- The code example involving the “name” variable should be on the board. Pupils will be asked to guess what they think the code will do.
- [Optional] Pupils can open their code editors and run the name example for themselves to get a sense of how it works.
- 4 VPC code examples are provided on slides 18-21. These can be done with the class to illustrate variables without the use of “input”.

Coding Activity (30 minutes)

- Pupils should implement a variable in their conversation programs. Pupils should be encouraged to modify their programs from lesson 2 to include this. If they are struggling to do this, encourage them to first explore the given example using the “name” variable and to modify the output line so that they are still using the name variable but the rest of the printed statement is different. From there, pupils can build up to modifying the rest of their program to use a different variable.

Closing (10 minutes)

- Pupils see examples of modern applications using variables through welcome messages such as “Welcome {name}!”. Pupils should be shown the on-screen images and encouraged to spot the variable for themselves.
- If there is time, it would be a good experience for pupils to find their own examples of these messages. If your school uses Glow or a similar platform there should be examples here that pupils can be directed to.

Spatial Skills

- Go through the VPC process when showing pupils the code examples.
- Abstracting variables from being shown programs in the closing activity is an opportunity to train spatial skills. Ensure enough time (around 10 minutes) is left for pupils to engage in this.